IBMHR Attrition Analysis Project Report

1. Introduction

Employee attrition is a critical challenge for organizations as it leads to increased hiring costs, loss of talent, and reduced productivity.

This project analyzes employee data to identify key factors contributing to attrition and provide actionable insights.

2. Objectives

Calculate overall attrition rate

Identify key drivers of attrition

Analyze attrition across department, salary, job satisfaction,

Provide data-driven recommendations

3. Tools & Technologies Used

Python (Pandas, NumPy, Matplotlib, Seaborn)

SQL (MySQL / SQL Server)

Power BI (Dashboard & Visualization)

4. Dataset Collection

Dataset Source : Kaggle.com

5. Data Analysis in Python

Step 1: importing libraries and load dataset

```
[186]: import pandas as pd
       df = pd.read_csv("D:\Desktop\project\project 2\ibm_hr_analytics_attrition_dataset.csv")
```

```
[187]: pd.set_option("display.max_columns", None)
       pd.set_option("display.max_rows", None)
       df.head(200)
```

[187]:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSa
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	
5	32	No	Travel_Frequently	1005	Research &	2	2	Life Sciences	1	8	

Step 2 : Understanding Data

```
]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   age                        1470 non-null   int64
 1   attrition                  1470 non-null   object
 2   business_travel            1470 non-null   object
 3   daily_rate                 1470 non-null   int64
 4   department                 1470 non-null   object
 5   distance_from_home         1470 non-null   int64
 6   education                  1470 non-null   int64
 7   education_field            1470 non-null   object
 8   employee_count             1470 non-null   int64
 9   employee_number            1470 non-null   int64
 10  environment_satisfaction   1470 non-null   int64
 11  gender                     1470 non-null   object
 12  hourly_rate                1470 non-null   int64
 13  job_involvement            1470 non-null   int64
 14  job_level                  1470 non-null   int64
 15  job_role                   1470 non-null   object
 16  job_satisfaction           1470 non-null   int64
 17  marital_status             1470 non-null   object
 18  monthly_income             1470 non-null   int64
 19  monthly_rate               1470 non-null   int64
 20  num_companies_worked       1470 non-null   int64
 21  over18                     1470 non-null   object
 22  over_time                  1470 non-null   object
 23  percent_salary_hike        1470 non-null   int64
 24  performance_rating         1470 non-null   int64
 25  relationship_satisfaction  1470 non-null   int64
 26  standard_hours             1470 non-null   int64
 27  stock_option_level         1470 non-null   int64
 28  total_working_years        1470 non-null   int64
 29  training_times_last_year   1470 non-null   int64
 30  work_life_balance          1470 non-null   int64
 31  years_at_company           1470 non-null   int64
 32  years_in_current_role      1470 non-null   int64
 33  years_since_last_promotion 1470 non-null   int64
 34  years_with_curr_manager    1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

```
]: df.shape
```

```
]: (1470, 35)
```

```
]: df.columns
```

```
]: Index(['age', 'attrition', 'business_travel', 'daily_rate', 'department',
       'distance_from_home', 'education', 'education_field', 'employee_count',
       'employee_number', 'environment_satisfaction', 'gender', 'hourly_rate',
       'job_involvement', 'job_level', 'job_role', 'job_satisfaction',
       'marital_status', 'monthly_income', 'monthly_rate',
       'num_companies_worked', 'over18', 'over_time', 'percent_salary_hike',
       'performance_rating', 'relationship_satisfaction', 'standard_hours',
       'stock_option_level', 'total_working_years', 'training_times_last_year',
       'work_life_balance', 'years_at_company', 'years_in_current_role',
       'years_since_last_promotion', 'years_with_curr_manager'],
      dtype='object')
```

```
]: df.isna().sum()
```

Step 3: Standardizing text columns

```
# replacing CamelCase with snake_case
import re
df.columns = [re.sub(r'(?<!^)(?=[A-Z])', '_', col).replace(" ", "_").lower() for col in df.columns]
df.columns = df.columns.str.strip()
```

Step 4: Data cleaning

```
df.isna().sum()
```

```
age                          0
attrition                    0
business_travel              0
daily_rate                   0
department                   0
distance_from_home           0
education                    0
education_field              0
employee_count               0
employee_number              0
environment_satisfaction     0
gender                       0
hourly_rate                  0
job_involvement              0
job_level                    0
job_role                     0
job_satisfaction             0
marital_status               0
monthly_income               0
monthly_rate                 0
num_companies_worked         0
over18                       0
over_time                    0
percent_salary_hike          0
performance_rating           0
relationship_satisfaction    0
standard_hours               0
stock_option_level           0
total_working_years          0
training_times_last_year     0
work_life_balance            0
years_at_company             0
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df["department"] = df["department"].replace({"Research & Development" : "R&D","Human Resources" : "HR"})
df["department"]
```

```
0       Sales
1         R&D
2         R&D
3         R&D
4         R&D
5         R&D
6         R&D
7         R&D
8         R&D
9         R&D
10        R&D
11        R&D
12        R&D
13        R&D
14        R&D
15        R&D
16        R&D
17        R&D
```

6. Exploratory Data Analysis (Python)

```
# Exploratory analysis
# 1. Total employee count
len(df)
```

```
1470
```

```
# 2. Attrition count & percentage
df["attrition"].value_counts()
```

```
attrition
No     1233
Yes     237
Name: count, dtype: int64
```

```
(df["attrition"].value_counts(normalize=True) * 100).round(2)
```

```
attrition
No     83.88
Yes    16.12
Name: proportion, dtype: float64
```

```
# 3. Avg age of employees leaving
df[df["attrition"] == "Yes"]["age"].mean().round(0)
```

```
np.float64(34.0)
```

```
# 4. Attrition by Gender
df[df["attrition"] == "Yes"]["gender"].value_counts()
```

```
gender
Male      150
Female     87
Name: count, dtype: int64
```

```python
# 5. Attrition by department
df[df["attrition"]== "Yes"]["department"].value_counts()
```

```
department
R&D      133
Sales     92
HR        12
Name: count, dtype: int64
```

```python
# 6. Attrition by job role
df[df["attrition"] == "Yes"]["job_role"].value_counts()
```

```
job_role
Laboratory Technician        62
Sales Executive              57
Research Scientist           47
Sales Representative         33
Human Resources              12
Manufacturing Director       10
Healthcare Representative     9
Manager                       5
Research Director             2
Name: count, dtype: int64
```

```python
# 7. Attrition by education field
df[df["attrition"] == "Yes"]["education_field"].value_counts()
```

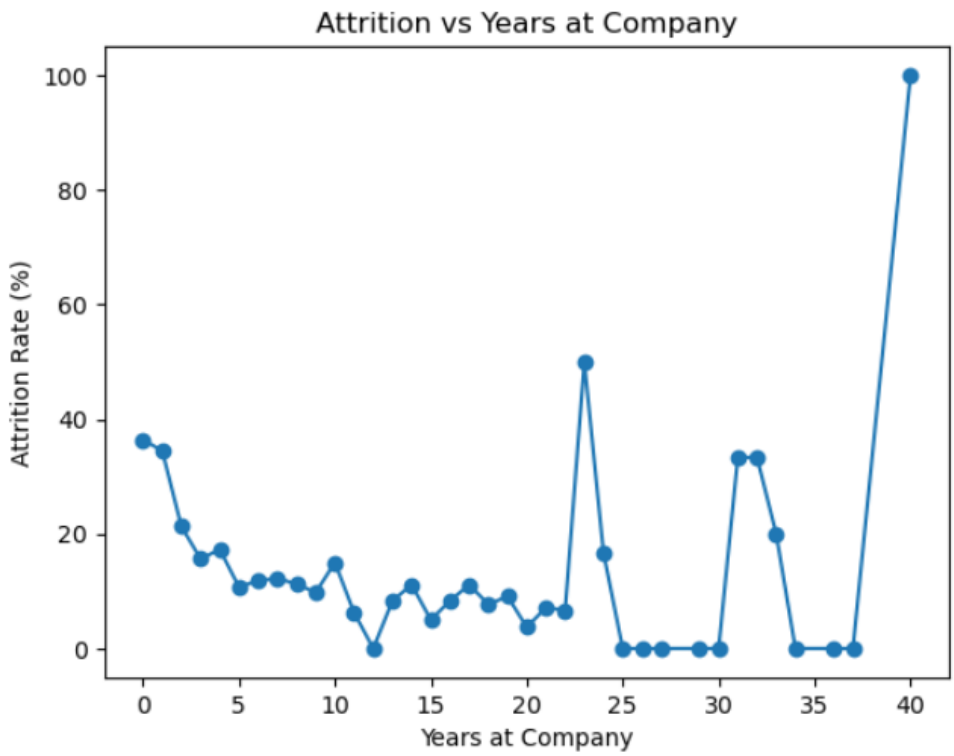
```
education_field
Life Sciences       89
Medical             63
Marketing           35
Technical Degree    32
Other               11
Human Resources      7
Name: count, dtype: int64
```

```python
# Advanced analysis
# 1.Attrition vs years at company
df["tenure_group"] = pd.cut(df["years_at_company"],
                            bins=[0,2,5,10,20,40],
                            labels=['0-2','2-5','5-10','10-20','20+'])
attrition_rate = (df["attrition"] == "Yes").groupby(df["tenure_group"]).mean() * 100
attrition_rate.round(2)
```

```
tenure_group
0-2       28.86
2-5       13.82
5-10      12.28
10-20      6.67
20+       12.12
Name: attrition, dtype: float64
```

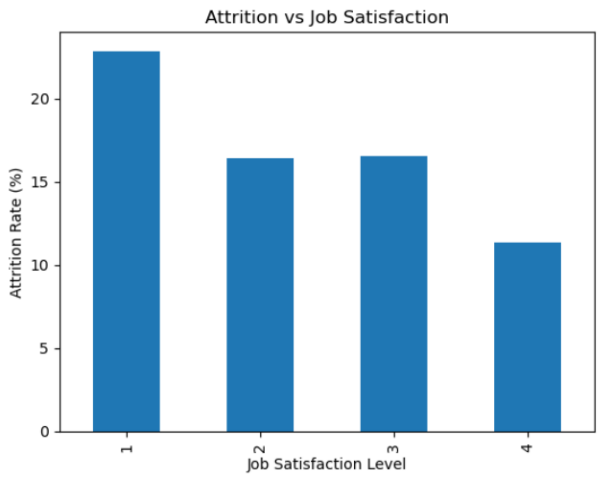
```python
# Trend analysis
import matplotlib.pyplot as plt
attrition_trend = (df["attrition"] == "Yes").groupby(df["years_at_company"]).mean() * 100
attrition_trend.plot(kind='line', marker='o')
plt.xlabel("Years at Company")      # X-axis label
plt.ylabel("Attrition Rate (%)")    # Y-axis label
plt.title("Attrition vs Years at Company")

plt.show()
```

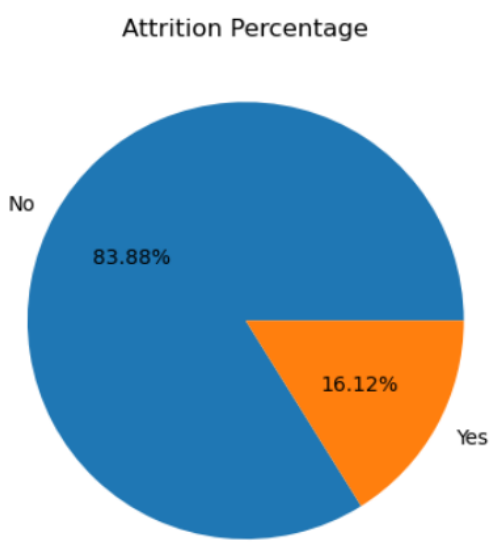


```python
# 2. Attrition vs job satisfaction

attrition_rate = (df["attrition"] == "Yes").groupby(df["job_satisfaction"]).mean() * 100
attrition_rate.plot(kind='bar')
plt.xlabel("Job Satisfaction Level")
plt.ylabel("Attrition Rate (%)")
plt.title("Attrition vs Job Satisfaction")
plt.show()
```

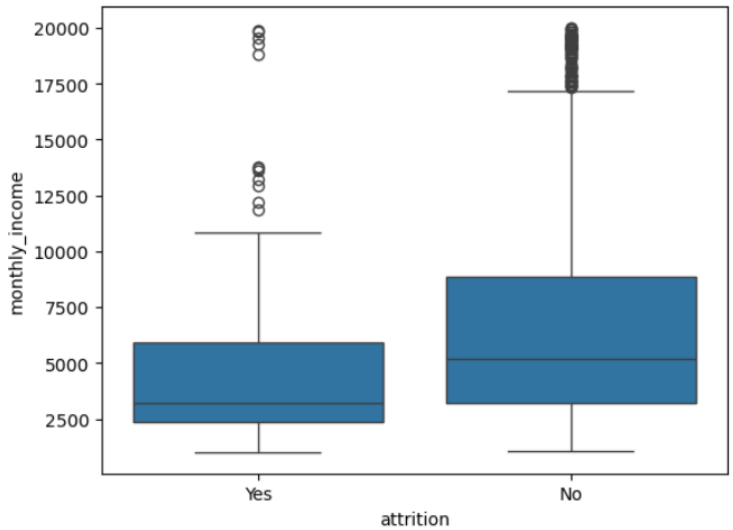



```python
# 3. Attrition rate
df["attrition"].value_counts().plot(kind='pie', autopct='%1.2f%%')
plt.title("Attrition Percentage")
plt.ylabel("")   # removes default label
plt.show()
```



```python
# 4.Boxplot (Salary vs Attrition)
import seaborn as sns
sns.boxplot(x ="attrition",y="monthly_income",data = df)
```

```
<Axes: xlabel='attrition', ylabel='monthly_income'>
```



6]:
```python
# procedure to connect DataFrame to SQL
!pip install sqlalchemy pymysql
from sqlalchemy import create_engine
engine = create_engine("mysql+pymysql://root:sushmiSa067$@localhost/hr_attrition")
df.to_sql("attrition",engine,if_exists = "replace",index = False)
```

```
Requirement already satisfied: sqlalchemy in c:\users\sushmitha\anaconda3\lib\site-packages (2.0.43)
Requirement already satisfied: pymysql in c:\users\sushmitha\anaconda3\lib\site-packages (1.1.2)
Requirement already satisfied: greenlet>=1 in c:\users\sushmitha\anaconda3\lib\site-packages (from sqlalchemy) (3.2.4)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\sushmitha\anaconda3\lib\site-packages (from sqlalchemy) (4.15.0)
```

6]: 1470

7]:
```python
df_check = pd.read_sql("SELECT * FROM attrition",engine)
df_check.head()
```

7]:

	age	attrition	business_travel	daily_rate	department	distance_from_home	education	education_field	employee_count	employee_number	environment_satisfacti
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	
1	49	No	Travel_Frequently	279	R&D	8	1	Life Sciences	1	2	
2	37	Yes	Travel_Rarely	1373	R&D	2	2	Other	1	4	
3	33	No	Travel_Frequently	1392	R&D	3	4	Life Sciences	1	5	
4	27	No	Travel_Rarely	591	R&D	2	1	Medical	1	7	

7. SQL Analysis

Step 1: Connected data from jupyter notebook

hr_attrition_analytics

```sql
CREATE DATABASE hr_attrition;
USE hr_attrition;
SELECT * FROM attrition;
```

Step 2: Basic queries

```sql
-- 1. Total number of employess
SELECT COUNT(*) as total_employees
FROM attrition;
```

total_employees
1470

```sql
-- 2. Count employees who left
SELECT COUNT(*) AS left_employee_count
FROM attrition
WHERE attrition = "Yes";
```

left_employee_count
237

```sql
-- 3. Attrition rate
SELECT ROUND(SUM(CASE WHEN Attrition = "Yes" THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS attrition_rate
FROM attrition;
```

attrition_rate
16.12

```sql
-- 4. Attrition by department
SELECT department,COUNT(*) AS attrition_count
FROM attrition
WHERE attrition = "Yes"
GROUP BY department
ORDER BY attrition_count DESC;
```

Result Grid | Filter Rows:

department	attrition_count
R&D	133
Sales	92
HR	12

```
-- 5. Average salary of employees who left
SELECT ROUND(AVG(monthly_income),0) as Avg_salary
FROM attrition
WHERE attrition = "Yes";
```

Result Grid

Avg_salary
4787

```
-- 6. Attrition by job role
SELECT job_role,COUNT(*) AS left_employees_count
FROM attrition
WHERE attrition = "Yes"
GROUP BY job_role
ORDER BY left_employees_count DESC;
```

Result Grid | Filter Rows:

job_role	left_employees_count
Laboratory Technician	62
Sales Executive	57
Research Scientist	47
Sales Representative	33
Human Resources	12
Manufacturing Director	10
Healthcare Representative	9
Manager	5
Research Director	2

```sql
-- 7. Employees with >5 years but left
SELECT COUNT(*) AS employees_left
FROM attrition
WHERE total_working_years > 5 AND attrition = "Yes";
```

Result Grid

	employees_left
▶	146

```sql
-- 8. Rank departments by attrition
SELECT department,COUNT(*) AS attrition_count,
DENSE_RANK() OVER (ORDER BY COUNT(*) DESC) AS rnk
FROM attrition
WHERE attrition = 'Yes'
GROUP BY department;
```

Result Grid Filter Rows:

	department	attrition_count	rnk
▶	R&D	133	1
	Sales	92	2
	HR	12	3

```sql
-- 9. Compare attrition vs satisfaction levels
SELECT job_satisfaction,
ROUND(SUM(CASE WHEN attrition = "Yes" THEN 1 ELSE 0 END) *100/COUNT(*),2) AS attrition_rate
FROM attrition
GROUP BY job_satisfaction
ORDER BY job_satisfaction;
```

Result Grid Filter Rows:

	job_satisfaction	attrition_rate
▶	1	22.84
	2	16.43
	3	16.52
	4	11.33

8. Power BI Dashboard

Step 1: Get data from SQL database and load it

Step 2: Created salary bucket column

```
Salary Bucket =
SWITCH(
    TRUE(),
    'hr_attrition_analysis'[monthly_income] < 2000, "<2K",
    'hr_attrition_analysis'[monthly_income] < 4000, "2K-4K",
    'hr_attrition_analysis'[monthly_income] < 6000, "4K-6K",
    'hr_attrition_analysis'[monthly_income] < 8000, "6K-8K",
    "8K+"
)
```

Step 3: Created measures for KPI's

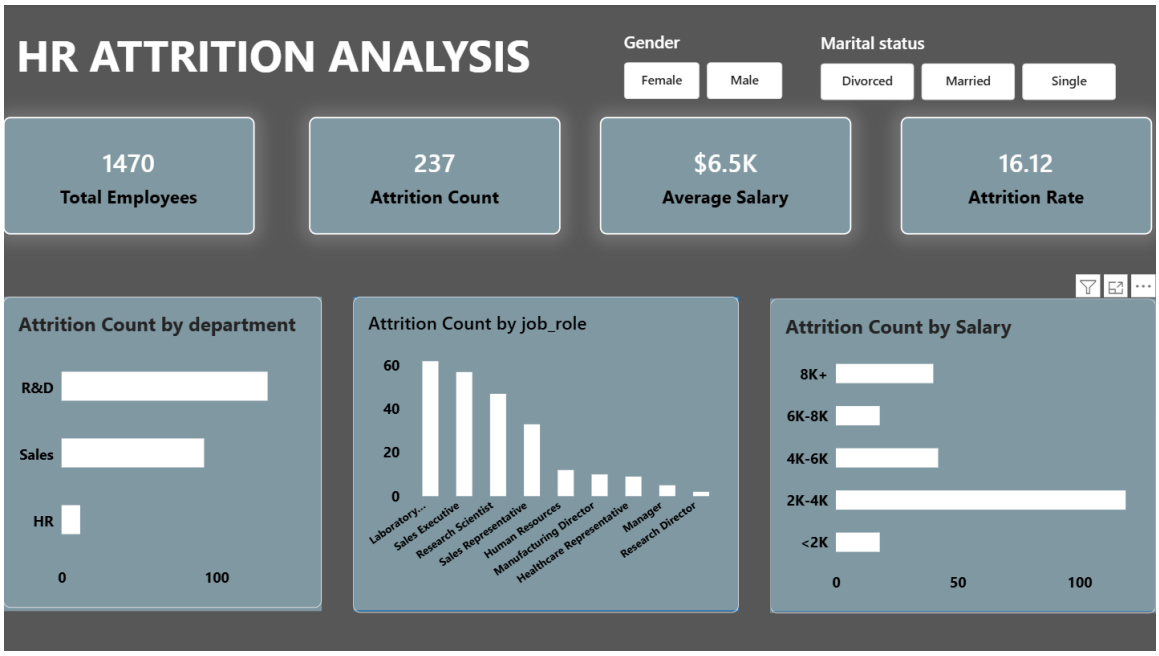
```
Attrition Count = CALCULATE(COUNTROWS('hr_attrition_analysis'),'hr_attrition_analysis'[attrition] = "Yes")
```

```
Attrition Rate = DIVIDE([Attrition Count], [Total Employees]) * 100
```

```
Average Salary = AVERAGE('hr_attrition_analysis'[monthly_income])
```

```
Total Employees = SUM('hr_attrition_analysis'[employee_count])
```

Step 3: Create dashboard with KPI's and charts



9. Key Insights

Low salary & low satisfaction → high attrition

Early tenure employees more likely to leave

Certain roles have higher attrition

10. Recommendations

Improve compensation

Enhance job satisfaction programs

Focus on early employee retention

Provide career growth

11. Conclusion

The project identifies key attrition drivers using Python, SQL, and Power BI.

Provides actionable business insights.

12. Future Scope

Build predictive model

Automate dashboards

Use real-time HR data